# R-practical

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Part A: Basic Operations

A1

```
v <- seq(5, 50, by = 5)
v
```

```
##  [1]  5 10 15 20 25 30 35 40 45 50
```

```
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2

```
w <- c(1,2,3)
rep(w, times = 4)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z <- c(1,2,3)
rep(w, each = 4)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
remainder <- 17 %% 5
integer_division <- 17 %/% 5
power <- 2 ^ 10
square_root <- sqrt(144)
pi_rounded <- round(pi, 3)

remainder
```

```
## [1] 2
```

```
integer_division
```

```
## [1] 3
```

```
power
```

```
## [1] 1024
```

```
square_root
```

```
## [1] 12
```

```
pi_rounded
```

```
## [1] 3.142
```

A4

```
a <- c(2,4,6,8)
b <- c(1,3,5,7)
a+b
```

```
## [1]  3  7 11 15
```

```
a*b
```

```
## [1]  2 12 30 56
```

```
a>b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
sum(a*b)
```

```
## [1] 100
```

A5

```
paste("Student", 1:3, sep ="-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```
typeof(42)
```

```
## [1] "double"
```

```
typeof(42L)
```

```
## [1] "integer"
```

```
typeof("42")
```

```
## [1] "character"
```

```
typeof(41i)
```

```
## [1] "complex"
```

```
typeof(TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Condition Statements

B1

```
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
```

```
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students[["average"]] <- round(mean((students[["final"]])) + mean((students[["midterm"]])),digits = 1)
students[["average"]]
```

```
## [1] 161.9 161.9 161.9 161.9 161.9 161.9 161.9 161.9
```

B4

```
students[["average"]] >= 70
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```
students[["attendance"]] >= 75
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

B5

```
students[["standing"]] <- ifelse(students[["average"]] >= 90, "Honors",
                          ifelse(students[["average"]] >= 75, "Good", "Needs Support"))

students[["standing"]]
```

```
## [1] "Honors" "Honors" "Honors" "Honors" "Honors" "Honors" "Honors" "Honors"
```

Part C

C1

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  gender = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80)
)
new_students[["average"]] <- (new_students[["midterm"]] + new_students[["final"]]) / 2
new_students[["passed"]] <- new_students[["average"]] >= 60
new_students[["standing"]] <- ifelse(new_students[["passed"]], "Pass", "Fail")

missing_cols <- setdiff(names(students), names(new_students))
for (col in missing_cols) new_students[[col]] <- NA
new_students <- new_students[, names(students), drop = FALSE]

students_all <- rbind(students, new_students)
```

C2

```
quiz <- c(8, 7, 10, 5, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
class(students_all)
```

```
## [1] "data.frame"
```

C3

```
m <- cbind(midterm = students_all[["midterm"]], final = students_all[["final"]])
m <- rbind(m, mean = colMeans(m))
```

C4

```
m <- cbind(midterm = students_all[["midterm"]], final = students_all[["final"]])
m <- rbind(m, mean = colMeans(m))
```

Part D

D1

```
students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2

```
students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3

```
students_all[students_all$attendance > 90, ]
```

```
##   id  name  sex section midterm final attendance average standing quiz
## 1  1 Alice    F       A      78    82         95   161.9   Honors    8
## 3  3  Chen    M       A      92    96        100   161.9   Honors   10
## 7  7 Grace    F       A      95    93         98   161.9   Honors   10
## 9  9  Ines <NA>       C      81    84         92    82.5     Pass    8
```

D4

```
#subset(students_all, (gender == "F" | sex == "F") & (section == "A" | section == "C") & final >= 75, s
```

D5

```
students_all$name[which.max(students_all$average)]
```

```
## [1] "Alice"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Jamal"
```

D6

```
sorted_students <- students_all[order(-students_all$average), ]
head(sorted_students, 3)
```

```
##   id  name sex section midterm final attendance average standing quiz
## 1  1 Alice   F       A      78    82         95   161.9   Honors    8
## 2  2 Bruno   M       B      85    79         88   161.9   Honors    7
## 3  3  Chen   M       A      92    96        100   161.9   Honors   10
```

E1

```
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)

levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

E2

```
table(students_all$sex)
```

```
##
## F M
## 4 4
```

```
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 1
##   M 1 2 1
```

e3

```
as.integer(students_all$sex)
```

```
##  [1]  1  2  2  1  1  2  1  2 NA NA
```

```
# The integers represent the internal numeric codes assigned to each factor level, mapping alphabetical
```

E4

```
students_all$sex_label <- factor(students_all$sex, levels = c("F", "M"), labels = c("Female", "Male"))

levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

E5

```
summary(students_all$sex_label)
```

```
## Female   Male    NAs
##      4      4      2
```

```
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] NA
```

```
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] NA
```

## R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

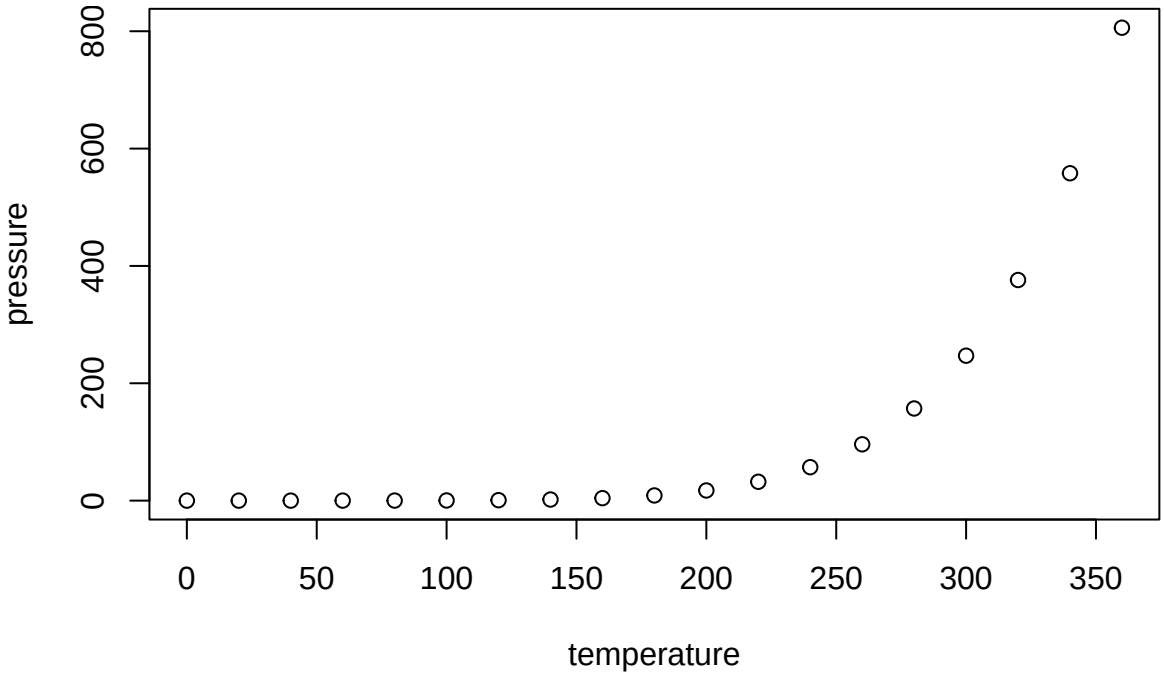
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00
```

## Including Plots

You can also embed plots, for example:




Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.